

Attention Mechanism Evolution: MHA → MQA → GQA

Source: [Vaswani et al., 2017] [Shazeer, 2019] [Ainslie et al., 2023]

MHA Multi-Head Attention

Q1 K1 V1
Q2 K2 V2
Q3 K3 V3
... 32 independent projections
KV Cache: $32 \times \text{head_dim}$

KV Cache: 100%
Quality: Best
Used in: GPT-3, BERT
100%
KV Cache baseline

MQA Multi-Query Attention

Q1 ─┐
Q2 ──┐ K V (shared)
Q3 ──┐
All heads share 1 set of KV
KV Cache: $1 \times \text{head_dim}$

KV Cache: 1/32
Quality: Slightly lower
Used in: Gemini
3%
KV Cache reduction

GQA Grouped-Query Attention

Q1-Q4 ─┐
Q5-Q8 ──┐ K1 V1
... ──┐
32 Q heads / 8 KV groups
KV Cache: $8 \times \text{head_dim}$

KV Cache: $8/32 = 25\%$
Quality: Near MHA
Used in: LLaMA 2/3
25%
KV Cache reduction